
Where is the planning tool's value mode-invariant?

think=off × think=on cross-mode aggregation of the two RQ decks · 5 PDDL tasks, ≥9B models · per-cell statistics only — raw trials are never pooled across arms or modes

Bottom line — one tool, three cross-mode regimes

Pairing every think=off cell with its think=on twin splits the five tasks into three regimes:

- ▶ solve — ROBUST. Realizable benefit $\geq +46$ pp in BOTH modes for every ≥ 9 B model. The gap shrinks under think=on only because reasoning rescues the unaided baseline (9B 11→27%, 35B 9→38%), not because the tool weakens (35B steered: 92% in both modes).
- ▶ simulate — SOLE-SOURCE. The baseline is 0% in both modes; the tool delivers +83...+97pp. No budget choice can confound a benefit that has no baseline to move.
- ▶ validate_domain / problem / plan — BUDGET-DEPENDENT. The dramatic think=on gaps (Δ up to +67pp vs off) appear exactly where the no-tools baseline truncates 78–100% and collapses. The honest mode-invariant claim is the robust floor: validate_plan +5...+16pp, validate_problem +20...+25pp, validate_domain +21pp (Gemma) to +74pp (9B, whose off baseline is already weak).

→ The verdict pattern — RQ0.1-0.4 YES/YES/MIXED/YES, RQ0.5 YES, RQ0.6 NO — reproduces in both modes (asserted at build time). The MAGNITUDES do not: they move with the decode budget.

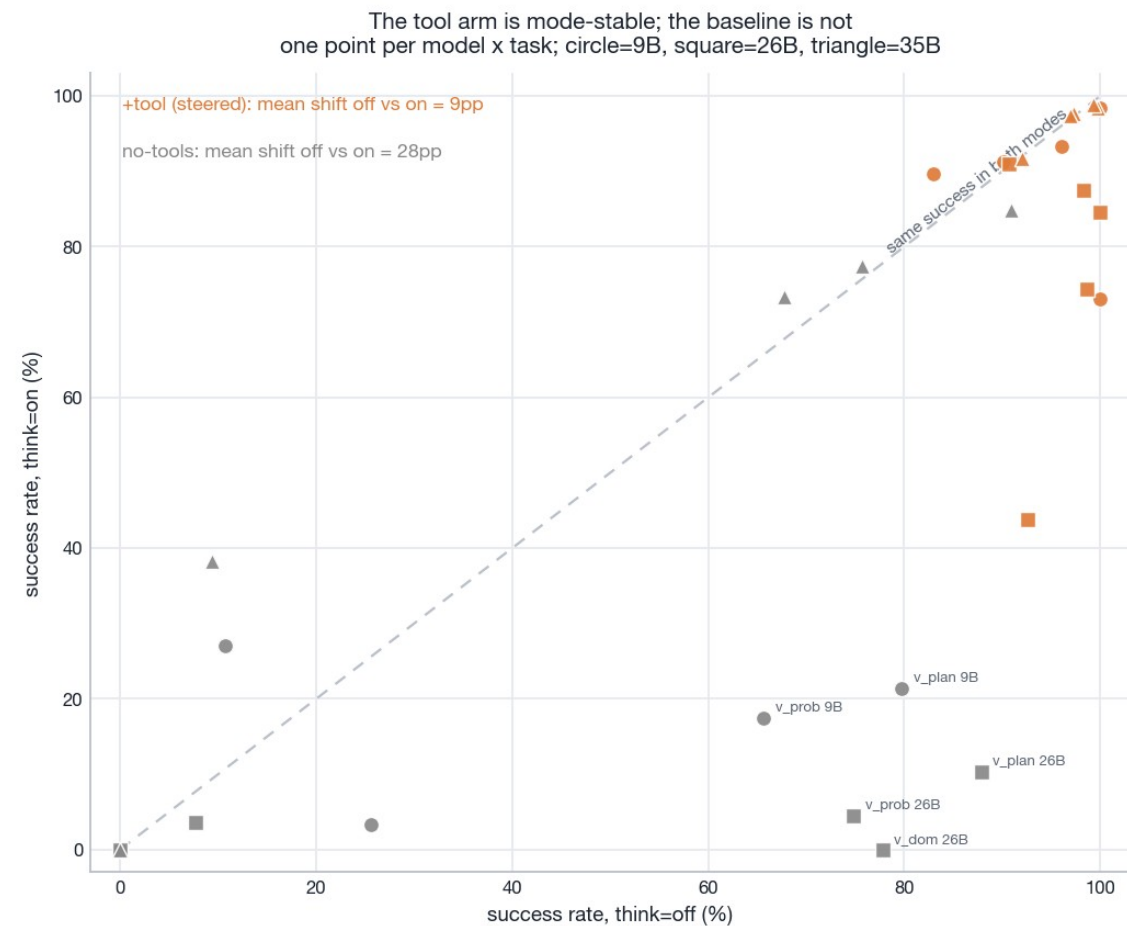
How this deck aggregates — and what it never does

Both source decks stay untouched: think=off is the locked headline, think=on the budget-confounded companion. This deck combines them only at the level of per-cell statistics.

- ▶ Spine metric: REALIZABLE benefit = success(+tool steered) – success(no-tools), per model × task × mode. Why the steered arm and not plain availability: two slides down.
- ▶ CI on each gap: Newcombe MOVER (Wilson-consistent difference of two independent proportions). CI on $\Delta(\text{on-off})$: MOVER again on the two gaps — valid because the off and on corpora share no trials. '*' = 95% CI excludes 0.
- ▶ Robust floor = min over modes of the realizable benefit — the budget-insensitive lower bound on what the tool is worth. Classes: robust (every $\geq 9\text{B}$ floor $\geq +30\text{pp}$) · sole-source (baseline floored in both modes) · budget-dep (some floor $< +30\text{pp}$).
- ▶ Raw trials are NEVER pooled across arms or think modes — each statistic comes from exactly one (model, mode, arm) corpus, and only the statistics are differenced or compared.

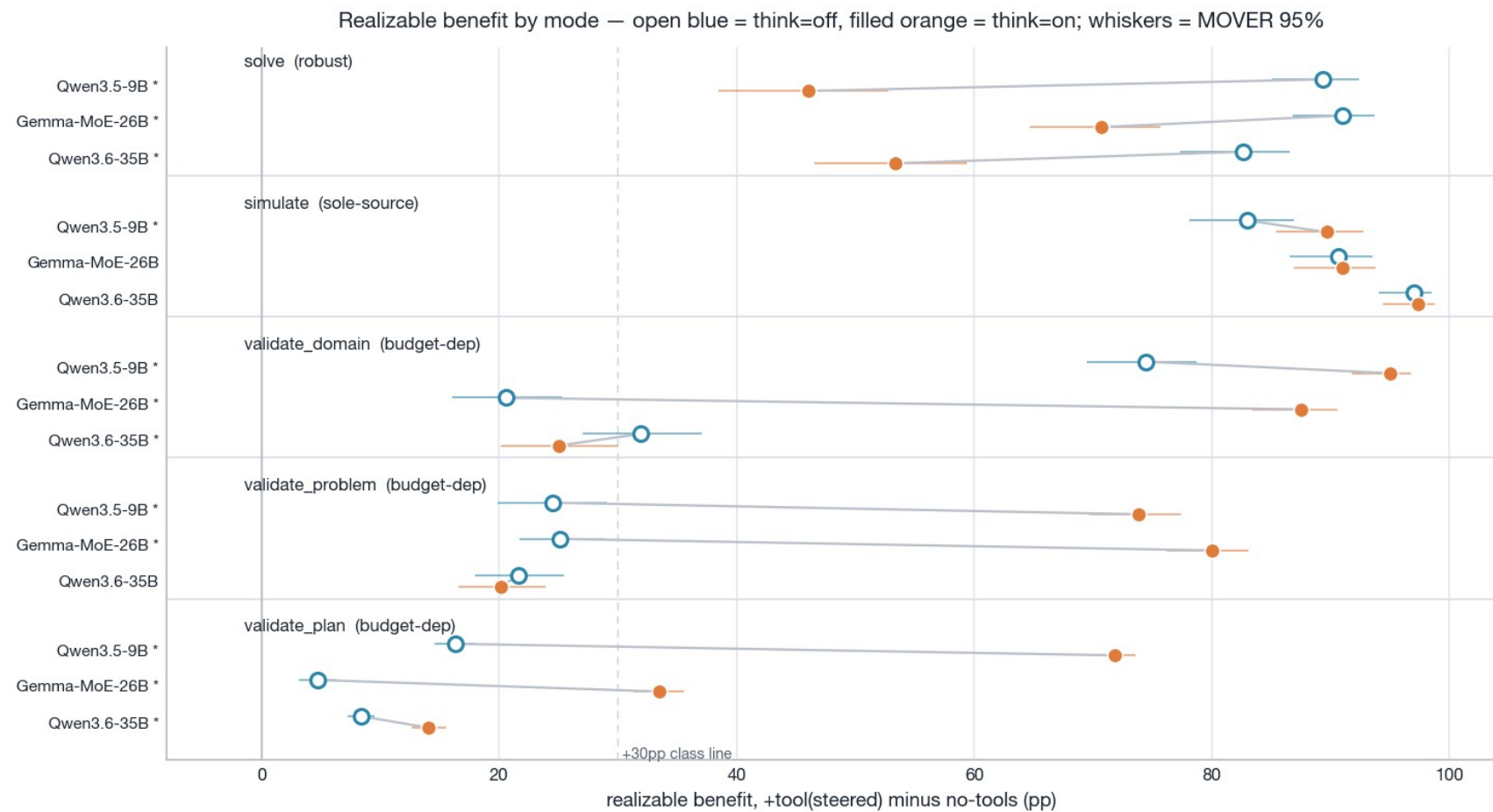
→ Standing caveats inherited from think=on: reasoning + answer share one 8,192-token decode budget (truncation differs by arm and mode — evidence slide near the end); completion tokens carry no logged thinking/answer split; latency is only defensible as turns × output tokens.

The gap moves because the BASELINE moves — not the tool arm



Success at think=off (x) vs think=on (y), one point per $\geq 9B$ model and task. The steered tool arm (orange) hugs the diagonal — mean shift 9pp, and Qwen3.6-35B's steered arm is fully mode-invariant on all five tasks. The no-tools baseline (grey) swings 28pp on average: down on the validation tasks (it reasons past the budget — Gemma validate_domain 78 $\rightarrow$ 0%, validate_plan 88 $\rightarrow$ 10%), up on solve (reasoning genuinely helps derivation). Every cross-mode gap change is driven by the grey points, so mode $\times$ arm interaction = baseline effect.

Realizable benefit by mode — what the floor protects



Benefit of the steered tool over no-tools, per ≥9B model: think=off (open blue) vs think=on (filled orange), MOVER 95% whiskers; '*' on the label = $\Delta(\text{on-off})$ significant (MOVER-D 95% CI excludes 0). solve stays large in both modes (floor +46pp); simulate is the tool or nothing in both; the validate_* dumbbells stretch right under think=on — that stretch is the budget artifact, and the open-blue end ($\approx$ the floor) is the defensible claim.

Cross-mode summary — five tasks, three classes (≥9B ranges)

task	realizable off (pp)	realizable on (pp)	robust floor (pp)	Δ(on-off), sig	class
solve	+83...+91	+46...+71	+46...+71	Δ -43...-20, 3/3	robust
simulate	+83...+97	+90...+97	+83...+97	Δ +0...+7, 1/3	sole-source
validate_domain	+21...+74	+25...+95	+21...+74	Δ -7...+67, 3/3	budget-dep
validate_problem	+22...+25	+20...+80	+20...+25	Δ -2...+55, 2/3	budget-dep
validate_plan	+5...+16	+14...+72	+5...+16	Δ +6...+56, 3/3	budget-dep

Per-model cross-mode detail — MOVER / MOVER-D 95% CIs • 1/2

task	model	realizable off	realizable on	$\Delta(\text{on-off})$ [95%]	floor
solve	Qwen3.5-9B	+89 [+85,+92]*	+46 [+39,+53]*	Δ -43 [-51,-35]*	+46pp
	Gemma-MoE-26B	+91 [+87,+94]*	+71 [+65,+76]*	Δ -20 [-27,-14]*	+71pp
	Qwen3.6-35B	+83 [+77,+86]*	+53 [+47,+59]*	Δ -29 [-37,-21]*	+53pp
simulate	Qwen3.5-9B	+83 [+78,+87]*	+90 [+86,+93]*	Δ +7 [+1,+12]*	+83pp
	Gemma-MoE-26B	+91 [+87,+93]*	+91 [+87,+94]*	Δ +0 [-5,+5]	+91pp
	Qwen3.6-35B	+97 [+94,+98]*	+97 [+95,+99]*	Δ +0 [-3,+4]	+97pp
validate_domain	Qwen3.5-9B	+74 [+70,+79]*	+95 [+92,+97]*	Δ +21 [+15,+26]*	+74pp
	Gemma-MoE-26B	+21 [+16,+25]*	+88 [+84,+91]*	Δ +67 [+61,+72]*	+21pp
	Qwen3.6-35B	+32 [+27,+37]*	+25 [+20,+30]*	Δ -7 [-14,-0]*	+25pp

Per-model cross-mode detail — MOVER / MOVER-D 95% CIs • 2/2

task	model	realizable off	realizable on	$\Delta(\text{on-off})$ [95%]	floor
validate_problem	Qwen3.5-9B	+24 [+20,+29]*	+74 [+70,+77]*	Δ +49 [+43,+55]*	+24pp
	Gemma-MoE-26B	+25 [+22,+29]*	+80 [+76,+83]*	Δ +55 [+50,+59]*	+25pp
	Qwen3.6-35B	+22 [+18,+25]*	+20 [+17,+24]*	Δ -2 [-7,+4]	+20pp
validate_plan	Qwen3.5-9B	+16 [+15,+18]*	+72 [+70,+74]*	Δ +56 [+53,+58]*	+16pp
	Gemma-MoE-26B	+5 [+3,+6]*	+33 [+31,+36]*	Δ +29 [+26,+31]*	+5pp
	Qwen3.6-35B	+8 [+7,+9]*	+14 [+13,+15]*	Δ +6 [+4,+7]*	+8pp

Why the spine is the STEERED arm, not plain availability

The per-mode decks lead with the availability gap (+tool plain vs no-tools, byte-identical wording). Across modes that gap stops measuring the tool:

- ▶ Under think=on the plain arm largely stops CALLING the tool — it reasons instead. solve tool_selected: Gemma 100→23%, 9B 100→59% (off→on); validate_plan: Gemma 21→1%.
- ▶ So a cross-mode availability comparison would conflate three things: the tool's value, the baseline's budget collapse, and a mode-dependent tool-calling failure that steering is known to repair (the off deck's RQ0.3 mechanism).
- ▶ The steered arm isolates 'what is the tool worth when actually invoked' — the quantity a deployment that prompts for tool use would realize. That is the only gap whose cross-mode difference is interpretable.
- ▶ Honesty note: steering does not fully de-confound Gemma — its steered validate_plan still only calls the tool 44% of the time under think=on (success 93→44% off→on), the one steered cell with a residual budget tax. 35B's steered arm is immune on all five tasks.

→ Availability and steering gaps remain fully reported, per mode, in the two source decks.

Cost-of-pass flips from a task regime to a model regime

task	model	off: no-tools	off: +tool steered	on: no-tools	on: +tool steered
solve	Qwen3.5-9B	46k	19k 0.4× cheaper	25k	26k 1.1× costlier
	Gemma-MoE-26B	50k	16k 0.3× cheaper	247k	22k 0.09× cheaper
	Qwen3.6-35B	50k	21k 0.4× cheaper	18k	19k 1.1× costlier
simulate	Qwen3.5-9B	— (0 succ)	24k only producer	— (0 succ)	20k only producer
	Gemma-MoE-26B	— (0 succ)	19k only producer	— (0 succ)	18k only producer
	Qwen3.6-35B	— (0 succ)	17k only producer	— (0 succ)	19k only producer
validate_domain	Qwen3.5-9B	9.3k	10k 1.1× costlier	200k	11k 0.06× cheaper
	Gemma-MoE-26B	1.9k	9.2k 4.9× costlier	— (0 succ)	12k only producer
	Qwen3.6-35B	3.1k	10k 3.2× costlier	6.0k	11k 1.8× costlier
validate_problem	Qwen3.5-9B	3.2k	15k 4.6× costlier	39k	15k 0.4× cheaper
	Gemma-MoE-26B	2.5k	11k 4.3× costlier	158k	14k 0.09× cheaper
	Qwen3.6-35B	2.8k	12k 4.4× costlier	4.1k	13k 3.3× costlier
validate_plan	Qwen3.5-9B	3.9k	14k 3.6× costlier	33k	16k 0.5× cheaper
	Gemma-MoE-26B	2.9k	13k 4.3× costlier	71k	27k 0.4× cheaper
	Qwen3.6-35B	3.7k	14k 3.8× costlier	6.0k	15k 2.5× costlier

The budget confound, cell by cell — truncation at the 8,192-token cap

task	model	no-tools off	no-tools on	steered off	steered on
solve	Qwen3.5-9B	20%	49%	35%	55%
	Gemma-MoE-26B	16%	92%	14%	35%
	Qwen3.6-35B	10%	24%	54%	31%
simulate	Qwen3.5-9B	37%	81%	74%	76%
	Gemma-MoE-26B	28%	95%	72%	75%
	Qwen3.6-35B	22%	73%	73%	78%
validate_domain	Qwen3.5-9B	10%	86%	0%	4%
	Gemma-MoE-26B	1%	100%	0%	12%
	Qwen3.6-35B	0%	9%	0%	1%
validate_problem	Qwen3.5-9B	5%	81%	1%	8%
	Gemma-MoE-26B	2%	96%	0%	18%
	Qwen3.6-35B	0%	0%	1%	4%
validate_plan	Qwen3.5-9B	2%	78%	10%	12%
	Gemma-MoE-26B	0%	90%	5%	52%
	Qwen3.6-35B	1%	11%	10%	10%

What the paper can claim — and how to say it

- ▶ Headline: report think=off as the clean, unconfounded read (locked verdicts YES/YES/MIXED/YES, RQ0.5 YES, RQ0.6 NO).
- ▶ Mode-invariance: the verdict PATTERN survives reasoning mode, and the robust floor is the quantified mode-invariant claim — solve $\geq +46$ pp, simulate $+83\dots+97$ pp (sole source), validate_problem $\approx +20\dots+25$ pp, validate_plan $\approx +5\dots+16$ pp, validate_domain $\geq +21$ pp.
- ▶ Budget-dependence: the large think=on availability/realizable gaps on the validation tasks are decode-budget artifacts (baseline truncation 78–100% on 9B/Gemma), NOT extra tool capability — cite the 35B control. Never quote a think=on validate_* gap without this caveat.
- ▶ What moves with mode (report as findings, not contradictions): RQ0.5's headroom case (validate_plan $+5 \rightarrow +27$ pp at off; solve $+48 \rightarrow +65$ pp at on); cost-of-pass regime (task-determined at off, model-determined at on); steering importance (grows under think=on — plain tool-calling collapses).

→ Open question (untested): whether a larger decode budget closes the think=on baseline gap — needs a cap-raised rerun; until then the floor is the claim that needs no such experiment.